



PROJECT PROPOSAL FOR KCA CONNECT AGENTIC AI

BIT 04104 FINAL YEAR PROJECT

**Program: BACHELOR OF SCIENCE IN
INFORMATION TECHNOLOGY**

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1. INTRODUCTION

1.1. Background

Higher learning institutions rely heavily on effective communication systems to serve students, staff, and administrators. At KCA University, information sharing is mainly done through emails, office visits, and group messaging platforms. While these methods provide basic communication, they are often slow, fragmented, and unreliable, especially when dealing with large student populations that require timely updates on academic and administrative matters.

Recent advances in Artificial Intelligence (AI) and conversational platforms present an opportunity to transform how institutions deliver information and services. Intelligent digital assistants powered by AI can provide real-time responses, automate administrative support, and create a centralized hub for academic and institutional services.

1.2. Problem Statement

KCA University and other higher-level institutions face challenges in delivering quick, personalized, and easily accessible information to students and staff. Current communication channels such as emails, physical visits, and group chats are inefficient and result in delays, missing information, and frustration among students. Learners often require immediate assistance with issues such as timetables, fee structures, unit registration, day to day consultation and exam information, needs that cannot be adequately met by the existing systems.

1.3. Proposed Solution

The proposed solution is KCA Connect Agentic AI, an AI-powered digital platform that acts as an intelligent assistant and central communication hub. The system will provide students, lecturers, and administrators with real-time, personalized, and accurate access to information and services. It will integrate academic and administrative data sources and offer features such as automated Q&A, instant notifications, and self-service tools.

1.4. Objectives

Main Objective

- ✓ To develop an AI-powered digital assistant for KCA University.

Specific Objectives

- ✓ To provide real-time access to academic and administrative information.

- ✓ To improve communication efficiency between students, lecturers, and administrators.
- ✓ To minimize delays and errors caused by manual and fragmented information delivery methods.

1.5. Justification

This project is necessary because students and staff need faster and more reliable access to information than current methods provide. An AI-powered system will reduce workload for administrators, improve decision-making for students, and enhance the overall user experience. It will also position KCA University as a forward-thinking institution that leverages modern technology to improve service delivery.

1.6. Scope

The scope of this project will focus on designing and developing the KCA Connect Agentic AI platform. The system will include:

- ✓ A conversational interface (chatbot/agent).
- ✓ Integration with key academic and administrative resources (e.g., timetables, fee structures, announcements).
- ✓ Real-time query response and notifications.
- ✓ Support for students, lecturers, and administrators.

The project will not initially include advanced predictive analytics or external third-party integrations, but it will be designed to scale for future features.

2. LITERATURE REVIEW

2.1. AI-Powered Communication Systems in Higher Institutions

Artificial Intelligence (AI) is increasingly being used in higher education to enhance communication, automate routine administrative processes, and provide personalized services. Universities across the world are adopting AI chatbots, virtual assistants, and smart portals to support student needs such as course registration, fee inquiries, timetables, and academic advising. AI-powered systems help reduce the workload of administrators while providing students with immediate, reliable, and personalized access to information (Holmes et al., 2019).

2.2. Existing Systems

2.2.1. Deakin University's "Genie" (Australia)

Deakin University introduced an AI-driven digital assistant called **Genie**. It provides students with real-time access to academic timetables, results, reminders, and university services. The system has improved efficiency in communication and increased student satisfaction by offering a 24/7 personalized assistant (Deakin University, 2018).

2.2.2. Georgia State University "Pounce"

Georgia State University implemented **Pounce**, an AI chatbot designed to guide students through admissions, enrolment, and financial aid processes. The system reduced "summer melt" (students failing to complete enrolment) by answering thousands of queries instantly. This demonstrated how AI systems can significantly improve student engagement and reduce administrative delays (Page & Gehlbach, 2017).

2.3. Case for KCA Connect

Unlike existing international systems such as Genie and Pounce, KCA Connect Agentic AI is tailored specifically for the KCA University. It addresses challenges at KCA University where students face delays due to reliance on emails, office visits, and group chats. KCA Connect will serve as a central hub for academic and administrative communication, enabling students to quickly access information such as timetables, fee structures, unit enquiries, and exam schedules. The information to be provided will be based on the campus branch selected e.g., main/town.

What makes KCA Connect unique is its ability to integrate seamlessly into KCA's existing systems and provide localized solutions tailored to Kenyan academic and administrative structures. It is not just an FAQ bot but an intelligent, agentic assistant capable of task execution, making it a more interactive and future-ready solution compared to existing manual systems.

2.4. Significance of the study

- ✓ **Insight from existing models:** The review provides practical insight into how AI assistants like Genie and Pounce were built and implemented in other universities, offering valuable guidance for developing KCA Connect.
- ✓ **Proof of concept:** It demonstrates that AI-powered systems are already improving communication, efficiency, and student satisfaction in higher education, validating the potential of KCA Connect.
- ✓ **Identification of gaps:** By comparing international systems to KCA's current methods, the review highlights the lack of localized, real-time, and centralized communication tools in Kenyan universities.
- ✓ **Design inspiration:** The success features of Genie and Pounce like real-time updates, personalized support, and automation serve as benchmarks for designing an effective and user-friendly AI system for KCA.
- ✓ **Institutional relevance:** The review emphasizes the need for an AI solution tailored to KCA's context, ensuring integration with existing systems and alignment with the unique academic and administrative structure of the university.

3. METHODOLOGY

3.1. Requirements Collection

To gather requirements for the KCA Connect Agentic AI system, the following methods will be used:

- ✓ **Interviews:** One-on-one discussions with students, lecturers, and administrators will provide deep insights into their pain points, preferences, and expectations. Interviews are chosen because they allow follow-up questions, clarification of issues, and collection of detailed qualitative data that may not emerge in written surveys.
- ✓ **Questionnaires:** Structured questionnaires will be distributed to a wider group of students and staff to collect quantifiable data on common needs such as access to timetables, fee inquiries, and administrative services. Questionnaires are selected because they are efficient, can reach a larger population quickly, and provide measurable statistics that can guide design priorities.

Using both interviews and questionnaires ensures a balanced approach — capturing both detailed personal experiences and broader statistical trends, leading to a more complete understanding of user requirements.

3.2. Development Methodology

The **Agile methodology** (Scrum framework) will be used for the development of KCA Connect Agentic AI. Agile is selected because:

- ✓ It supports iterative development, allowing the project to be built in small increments and tested frequently to validate functionality.
- ✓ It reduces risks of developing features that do not align with user expectations.
- ✓ It provides flexibility in adapting to changes in requirements as the project progresses.

The development cycle will involve short sprints, each delivering a functional component of the system (e.g., chatbot interface, timetable integration, notifications). This approach ensures the final system is both usable and relevant to the KCA University environment.

4. APPENDICES

4.1. Resources

Resource	Specification	Usage
Computer	8GB RAM, 500 SSD, core i7	To develop and store app
Internet connection	20 mbps	To connect to online resources
Flash Disk	32 GB	Back up storage
Programming Language	Python 3.8, HTML, Tailwind CSS, JavaScript	System Development
Printer	Epson 501 laser	Documentation
Development Languages	HTML, Tailwind CSS, JavaScript, Llama, LangChain framework, Python 3.8	System Development
Database	Firebase Qdrant	Login Authentications Store embeddings from KCA Data

4.2. Budget

Resource	Quantity	Availability	Estimated Cost (Ksh)	Actual Cost (Ksh)
Computer	1	Own	50000	46000
Internet connection	12 hours	Own	4000x6 =24000	4000x6 =24000
Flash Disk	1, 32GB	Own	1000	800
Printing Papers	1 rim	Own	800	600
Printer	1	Own	30000	25000
Total			105800	96000

4.3. Project Schedule

Activity	Expected start	Expected finish	Actual start	Actual Finish
Idea	5/10/2025	12/10/2025	6/10/2025	6/10/2025
Proposal presentation	9/10/2025	13/10/2025		
Proposal document	5/10/2025	12/10/2025	6/10/2025	10/10/2025
SRS (System Requirement specification)	13/10/2025	29/10/2025		
SDS (System design specification)	30/10/2025	18/11/2025		
System implementation	20/11/2025	2/12/2025		
Test plan	8/1/2026	23/1/2026		
User manual	24/1/2026	2/2/2026		
Final system presentation	14/3/2025	28/3/2026		
Compiled system	2/4/2026	4/4/2026		
Final report	5/4/2026	10/4/2026		

4.4. Project Risks and Mitigation

Risk	Effect	Mitigation
✓ Hardware Failure ✓ Laptop Theft ✓ System crash	Loss of data and progress	Use of backup devices to store the progress of the project.
Power disruption	Delayed operations	Use of UPS
Virus attacks	System Failure	Back up copy

4.5. References

- Deakin University. (2018). Genie digital assistant: Transforming student experience with AI. Retrieved from [<https://www.deakin.edu.au>]
- Holmes, W., Bialik, M., & Fadel, C. (2019). Artificial Intelligence in Education: Promises and Implications for Teaching and Learning. Boston, MA: Centre for Curriculum Redesign.
- Page, L. C., & Gehlbach, H. (2017). How Georgia State University used an AI chatbot to help students navigate enrolment. Education Next, 17(4), 68–74.